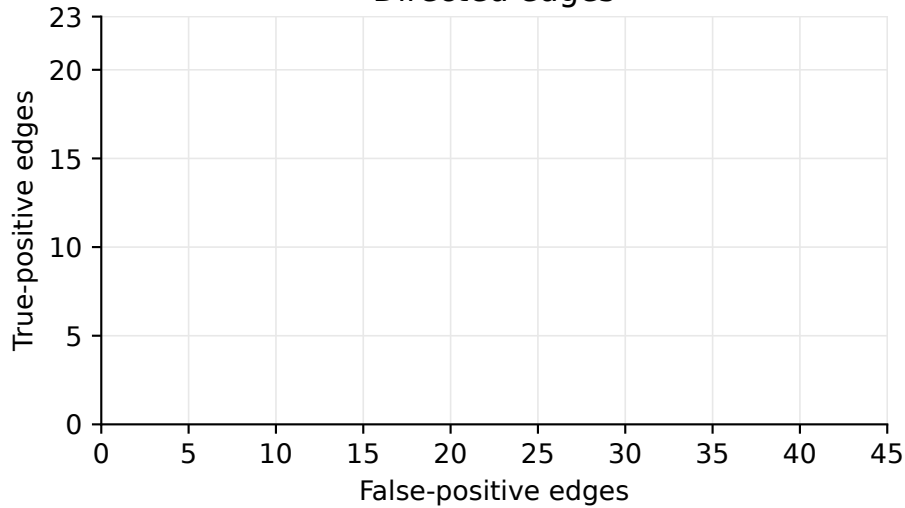


Causal Chambers: seven-condition pilot

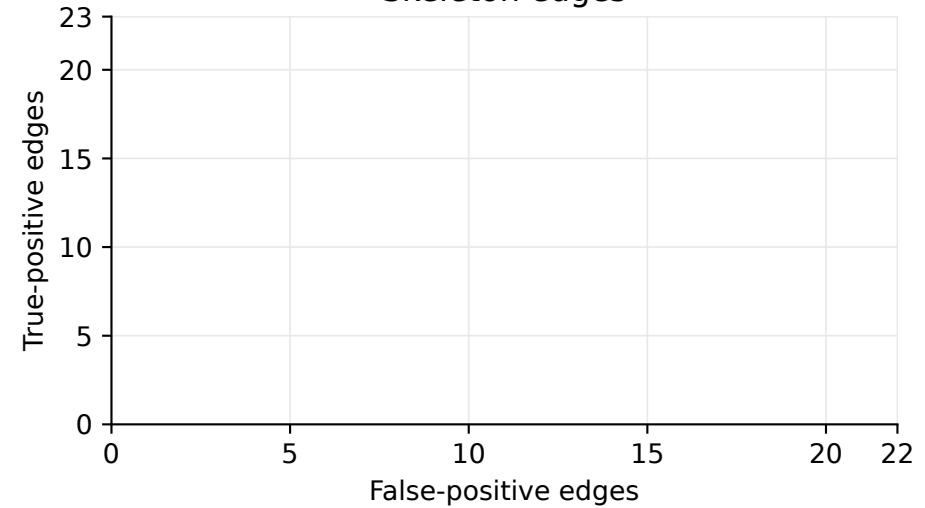
0/194 successful fits | 40,000 observations per configuration | no rank transformation

Actuator chain (scm_4)

Directed edges

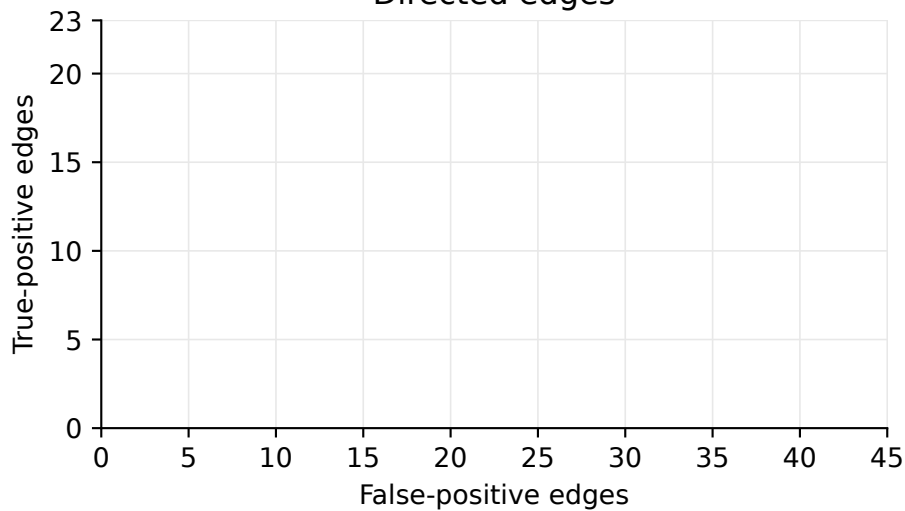


Skeleton edges

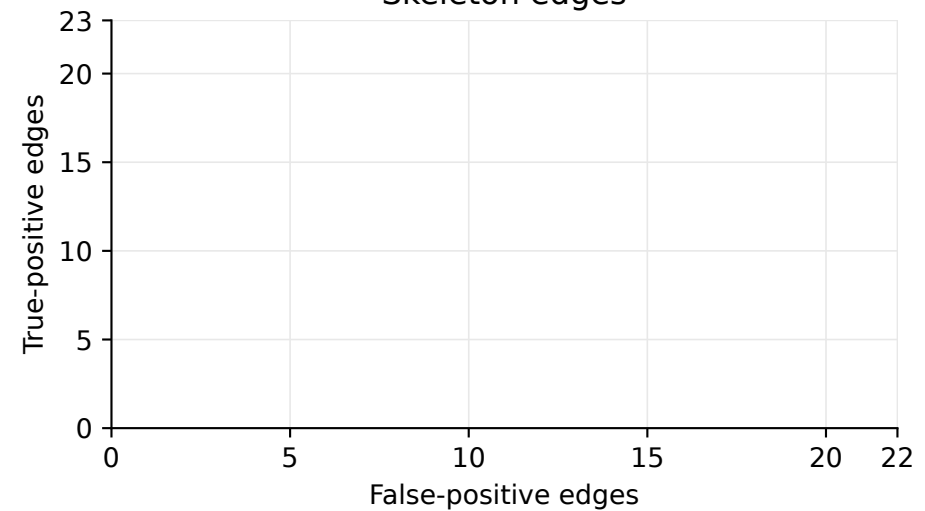


Actuator collider (scm_5)

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets
- ◇ PS-MIP: hidden targets (diagnostic)
- △ UT-IGSP: targets supplied
- ▽ GnIES: target union supplied
- ⊕ GIES: complete targets
- ⊗ IGSP: complete targets

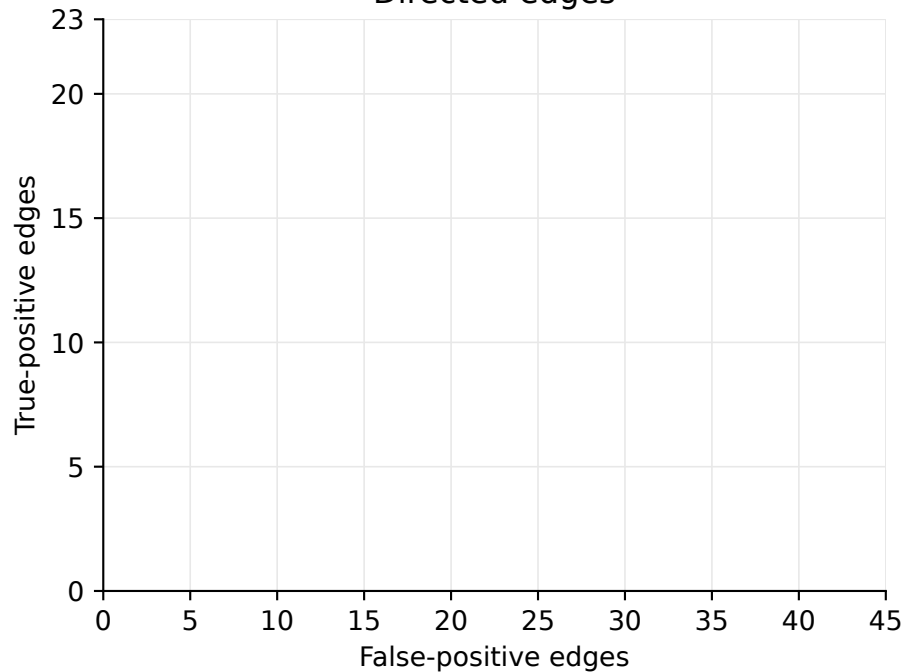
Reference: 21 physical + 2 programmed links. Full paths; no lines, envelope, or AUC.

Causal Chambers | Actuator chain (scm_4)

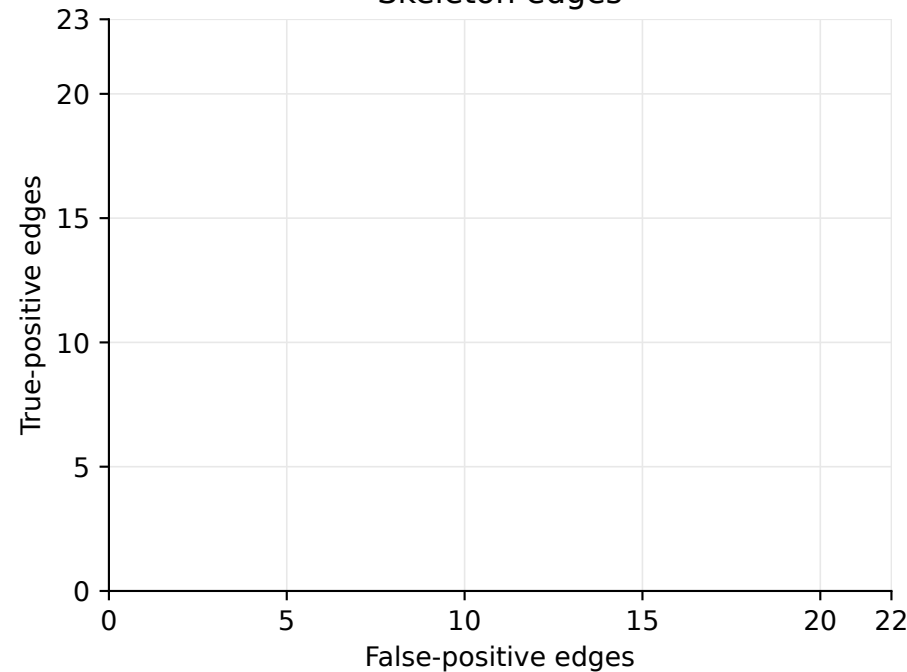
40,000 observations | 10 variables | 23 reference edges | 0/97 successful fits

Full-range paths; every successful operating point retained

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets
- ◇ PS-MIP: hidden targets (diagnostic)
- △ UT-IGSP: targets supplied
- ▽ GnIES: target union supplied
- ⊕ GIES: complete targets
- ⊗ IGSP: complete targets

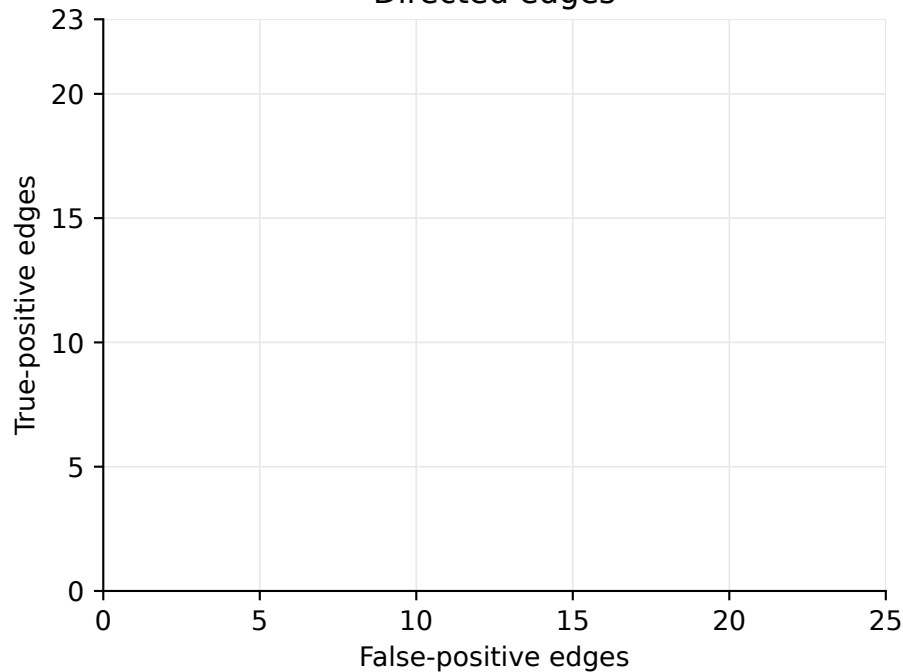
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator chain (scm_4)

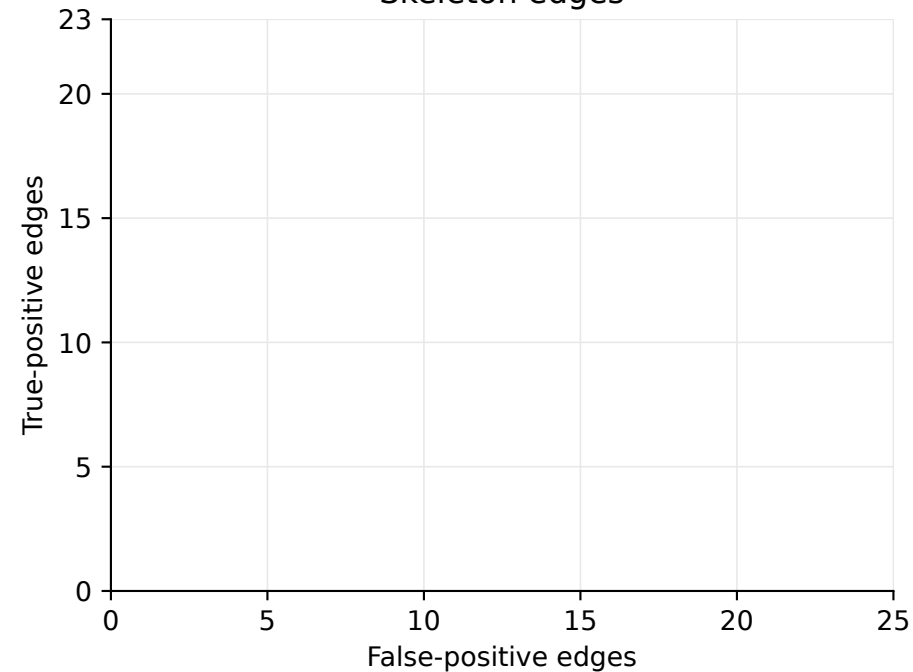
40,000 observations | 10 variables | 23 reference edges | 0/97 successful fits

0-25 FP viewing window; other points remain in full results

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets
- ◇ PS-MIP: hidden targets (diagnostic)
- △ UT-IGSP: targets supplied
- ▽ GnIES: target union supplied
- ⊕ GIES: complete targets
- ⊗ IGSP: complete targets

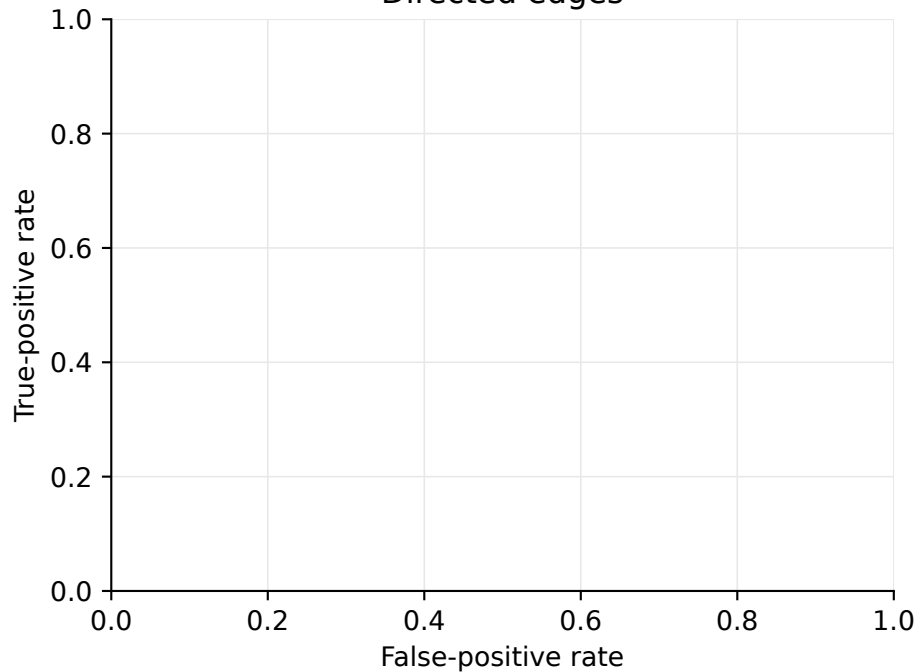
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator chain (scm_4)

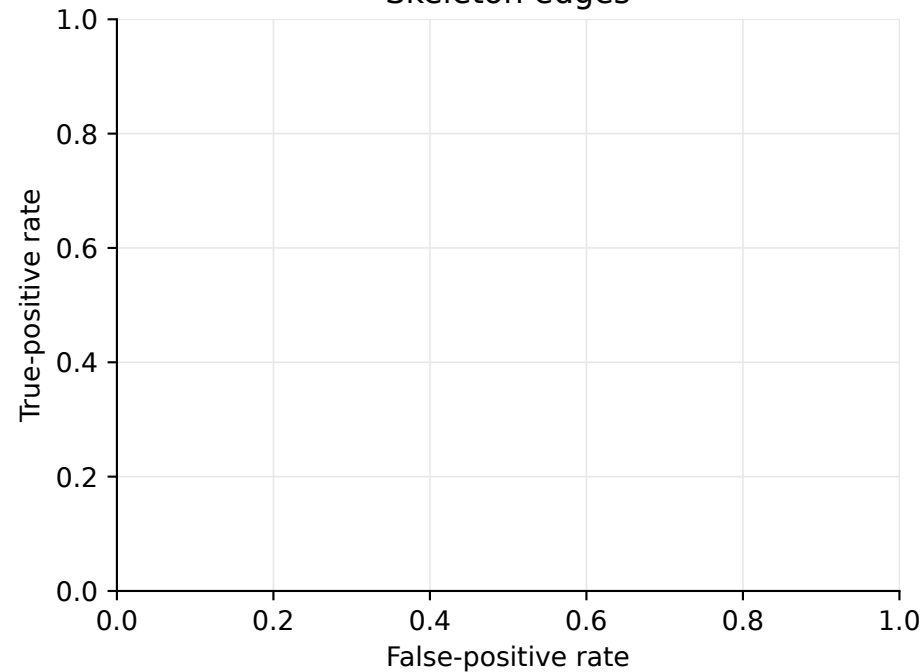
40,000 observations | 10 variables | 23 reference edges | 0/97 successful fits

Full-range normalized paths

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets
- ◇ PS-MIP: hidden targets (diagnostic)
- △ UT-IGSP: targets supplied
- ▽ GnIES: target union supplied
- ⊕ GIES: complete targets
- ⊗ IGSP: complete targets

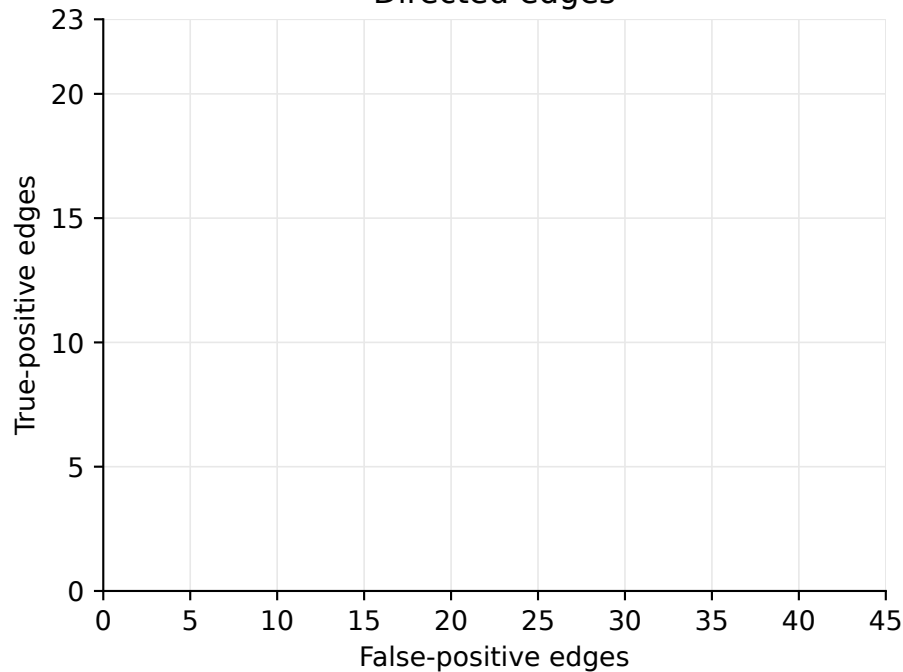
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator chain (scm_4)

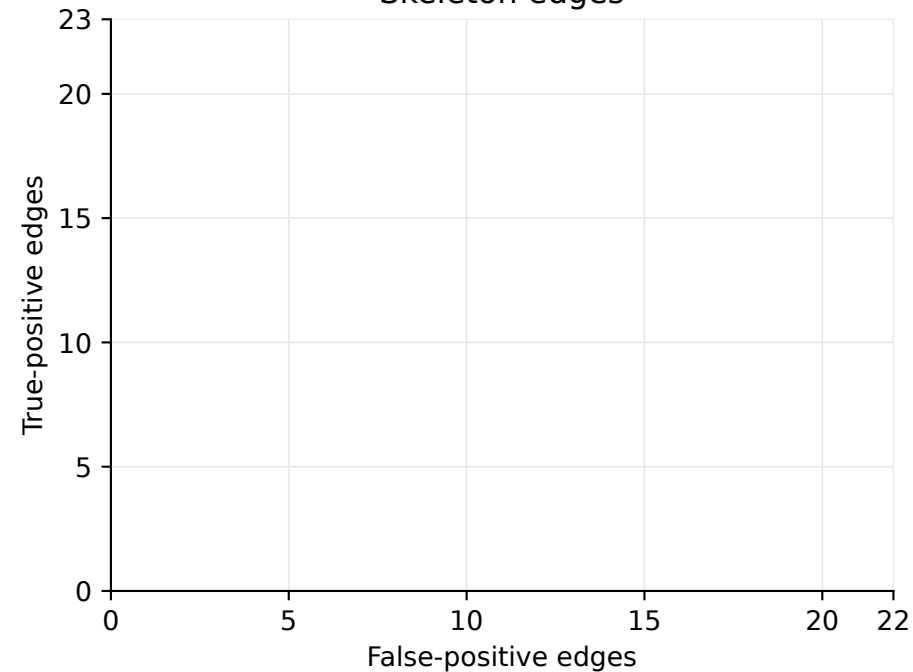
40,000 observations | 10 variables | 23 reference edges | 0/51 successful fits

Full-range paths; every successful operating point retained

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets

- ◇ PS-MIP: hidden targets (diagnostic)

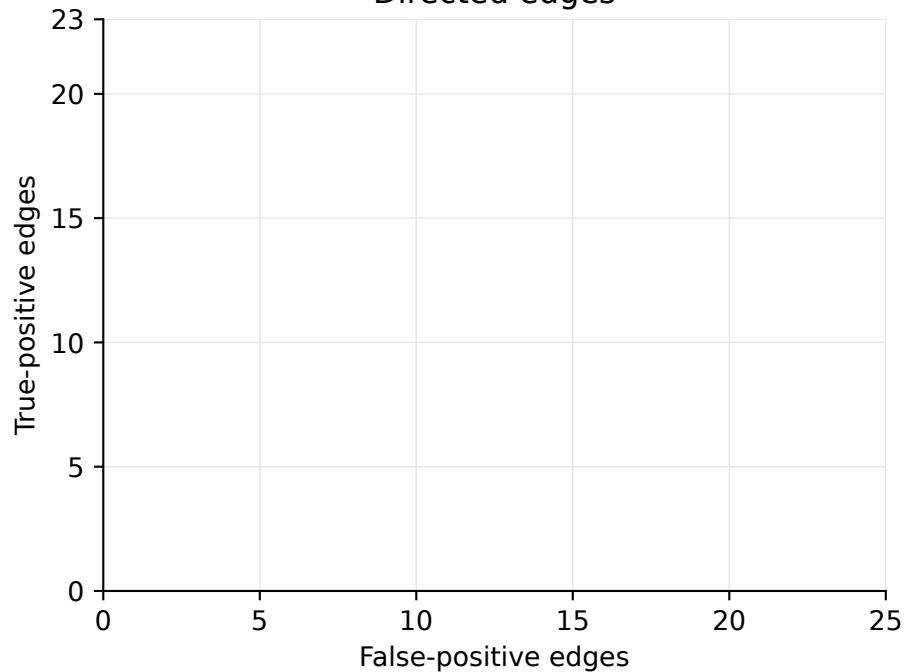
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator chain (scm_4)

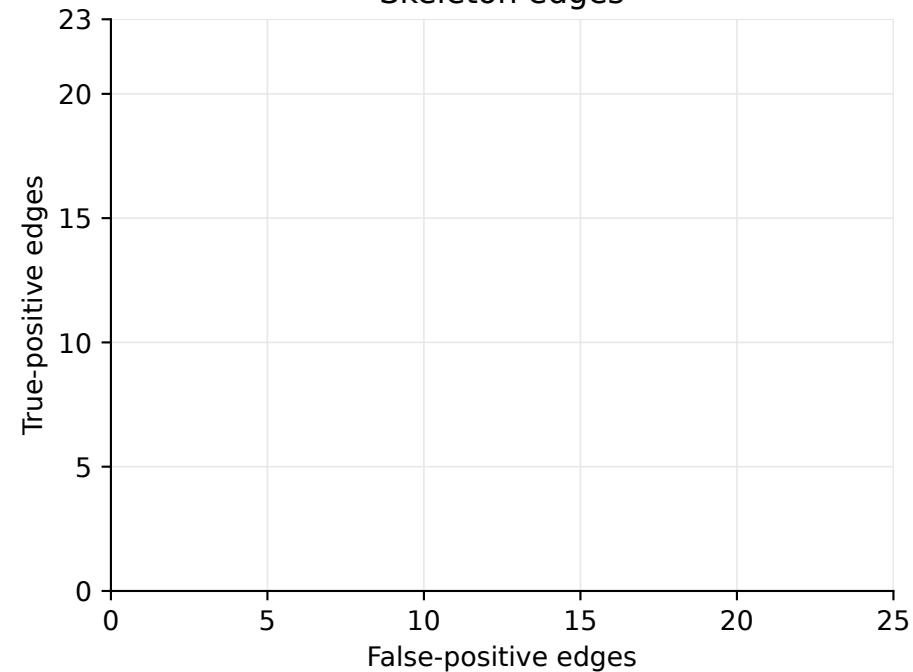
40,000 observations | 10 variables | 23 reference edges | 0/51 successful fits

0-25 FP viewing window; other points remain in full results

Directed edges



Skeleton edges



○ PS-MIP: complete targets
□ PS-MIP: known-present targets

◇ PS-MIP: hidden targets (diagnostic)

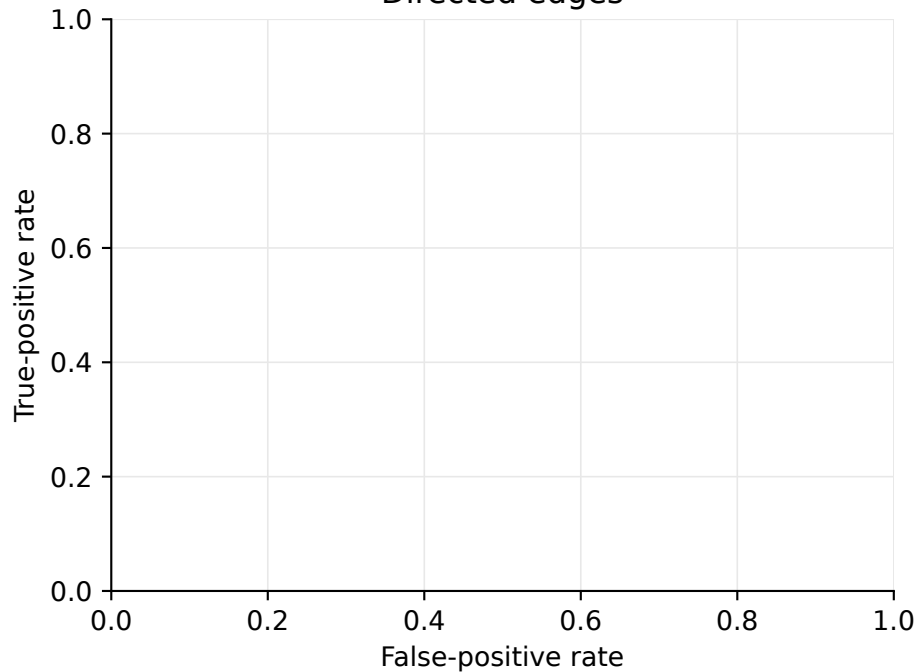
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator chain (scm_4)

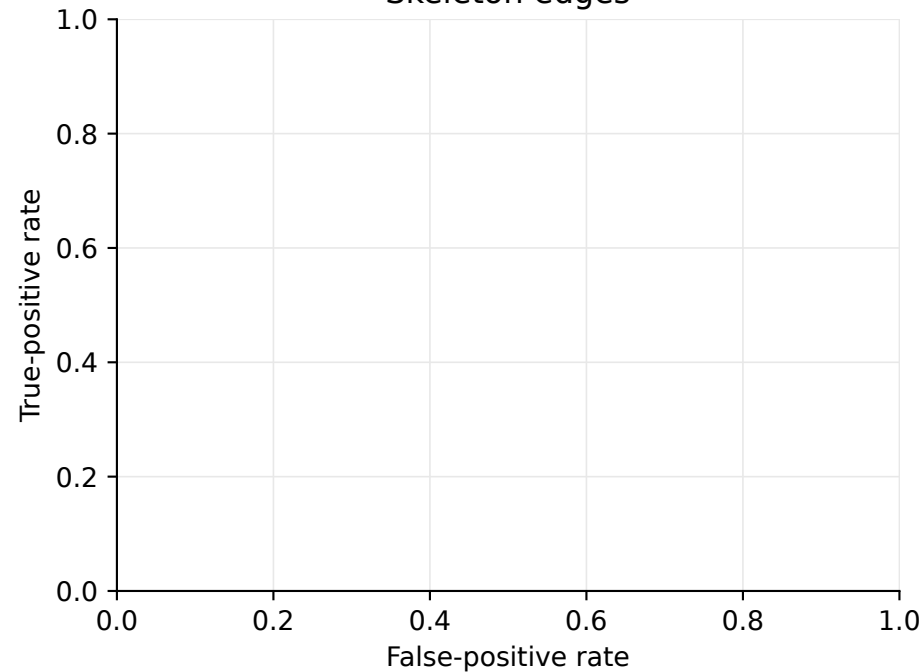
40,000 observations | 10 variables | 23 reference edges | 0/51 successful fits

Full-range normalized paths

Directed edges



Skeleton edges



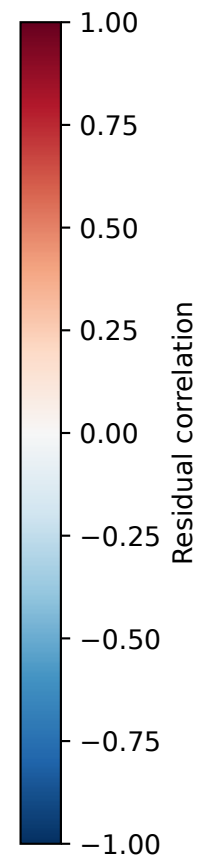
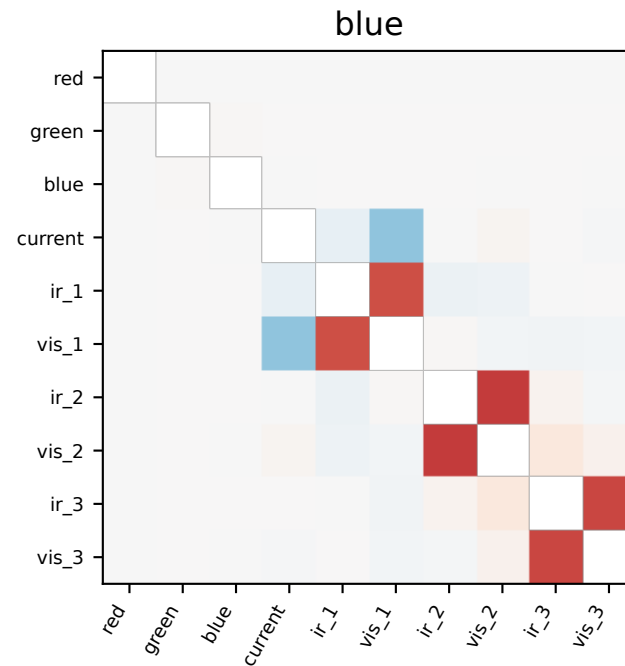
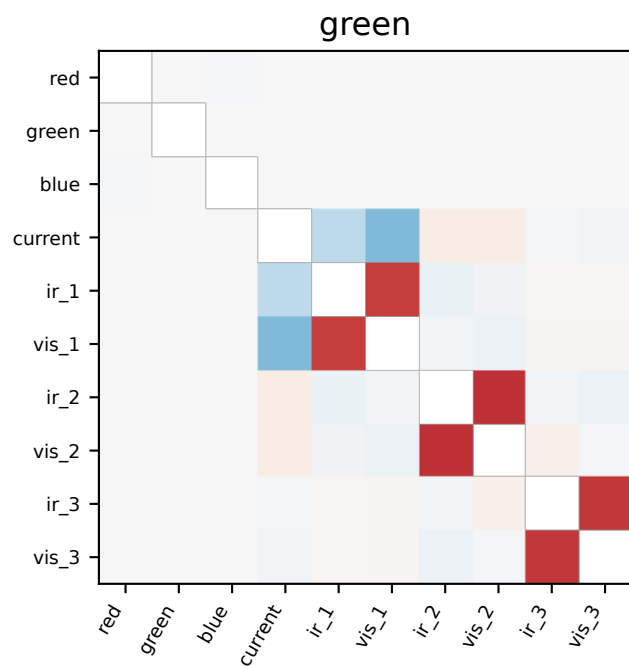
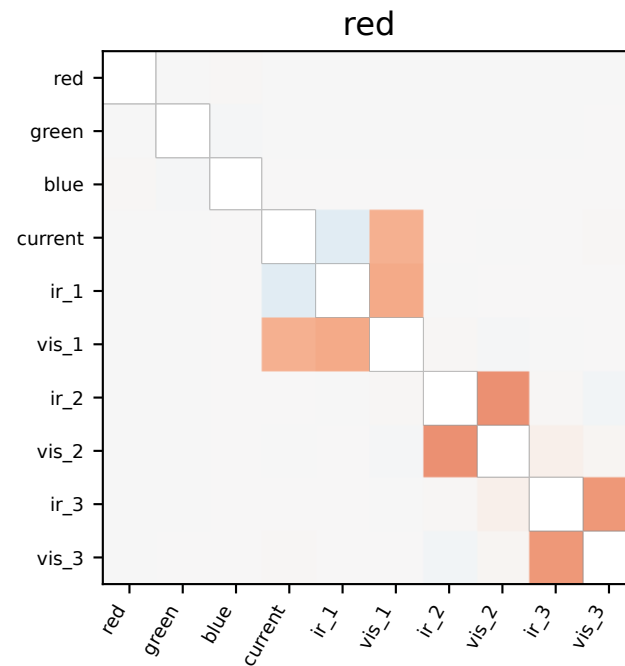
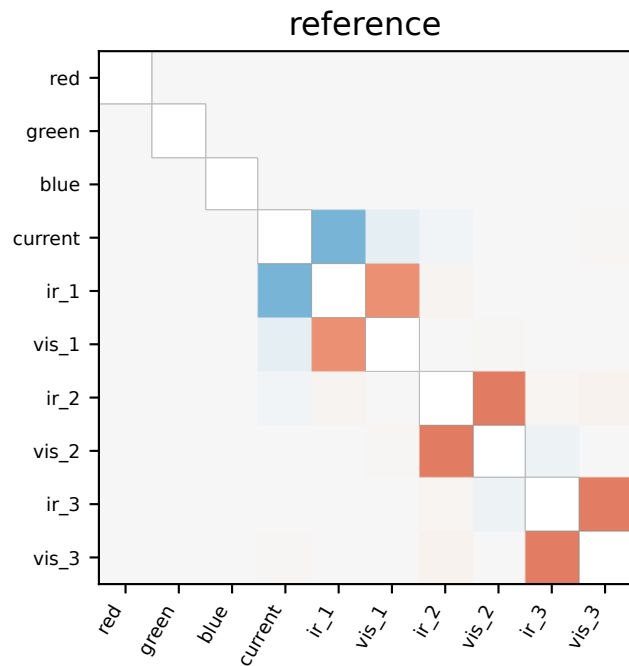
○ PS-MIP: complete targets
□ PS-MIP: known-present targets

◇ PS-MIP: hidden targets (diagnostic)

Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Model check: scm_4

Residuals after regression on documented parents, fitted separately within each environment



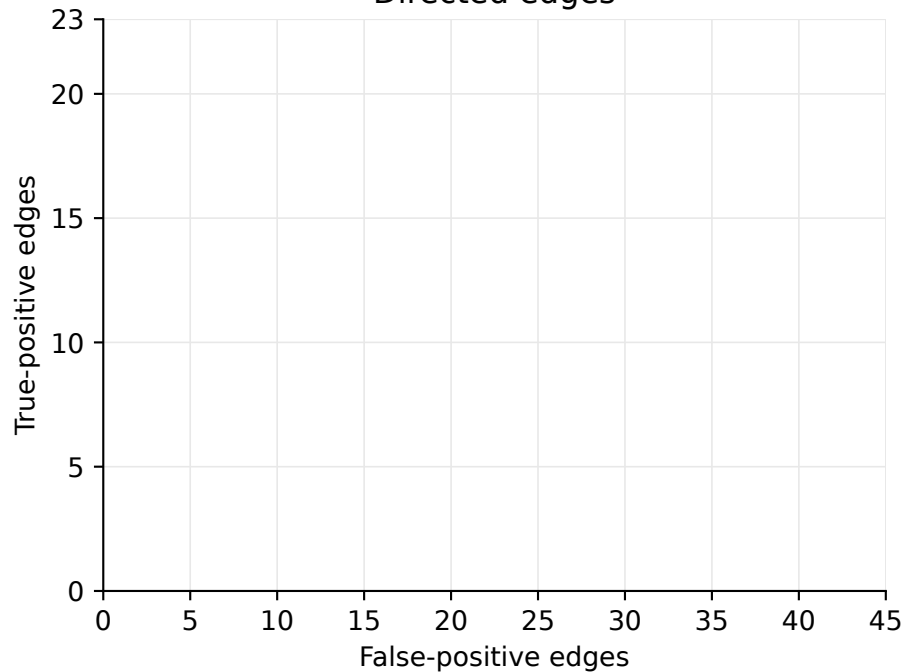
Off-diagonal dependence conflicts with independent SEM errors. No rows or variables were removed.
These diagnostics do not identify a unique cause or establish that omitted physical effects are absent.

Causal Chambers | Actuator collider (scm_5)

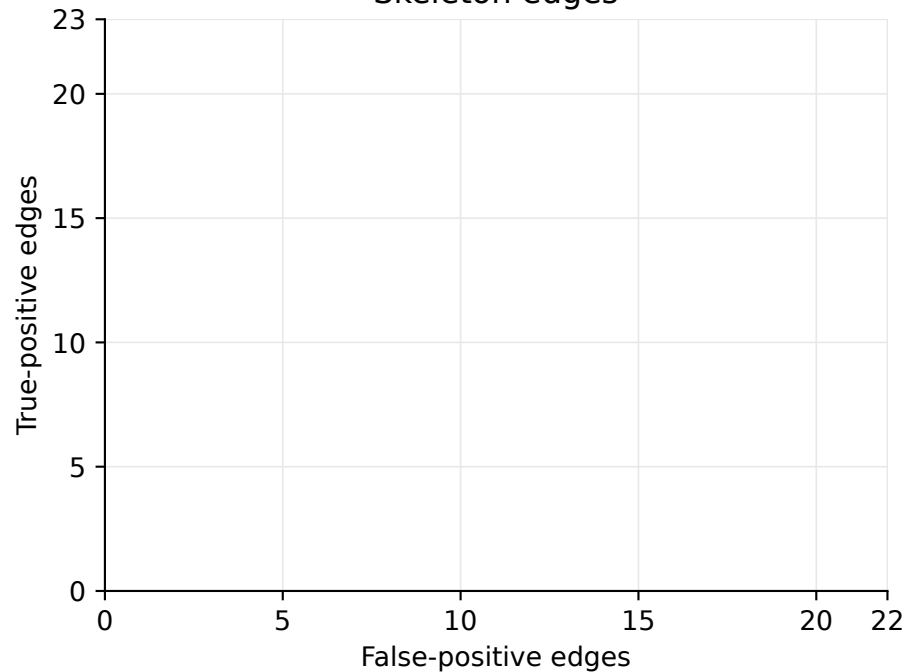
40,000 observations | 10 variables | 23 reference edges | 0/97 successful fits

Full-range paths; every successful operating point retained

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets
- ◇ PS-MIP: hidden targets (diagnostic)
- △ UT-IGSP: targets supplied
- ▽ GnIES: target union supplied
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- ⊗ IGSP: complete targets

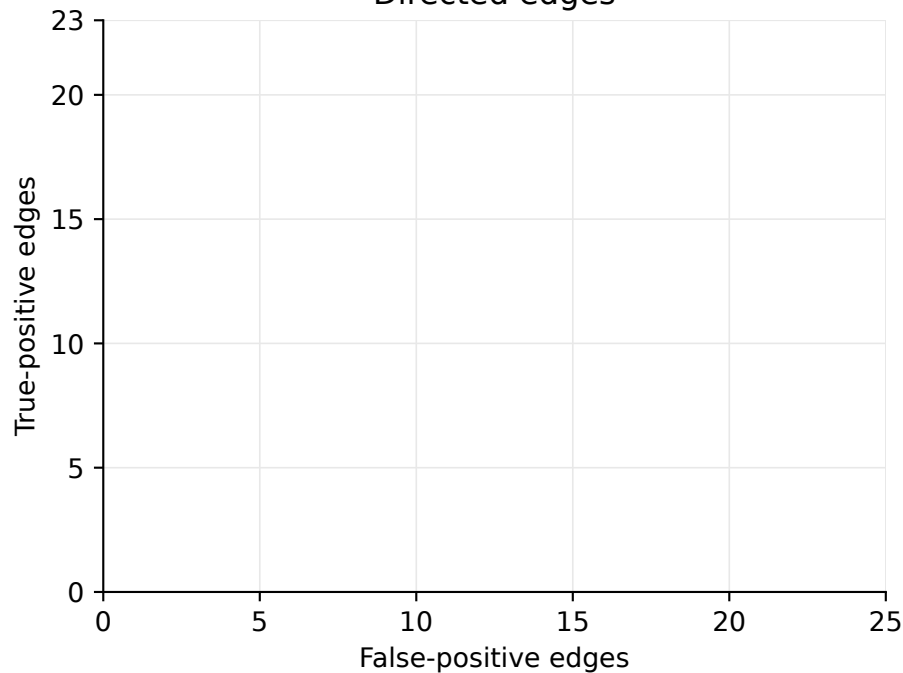
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator collider (scm_5)

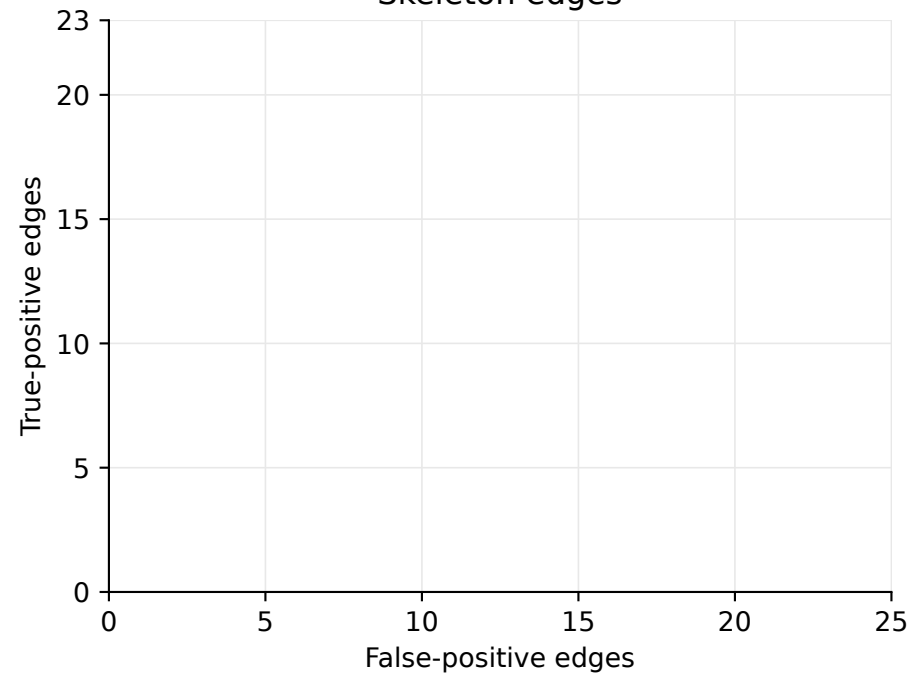
40,000 observations | 10 variables | 23 reference edges | 0/97 successful fits

0-25 FP viewing window; other points remain in full results

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets
- ◇ PS-MIP: hidden targets (diagnostic)
- △ UT-IGSP: targets supplied
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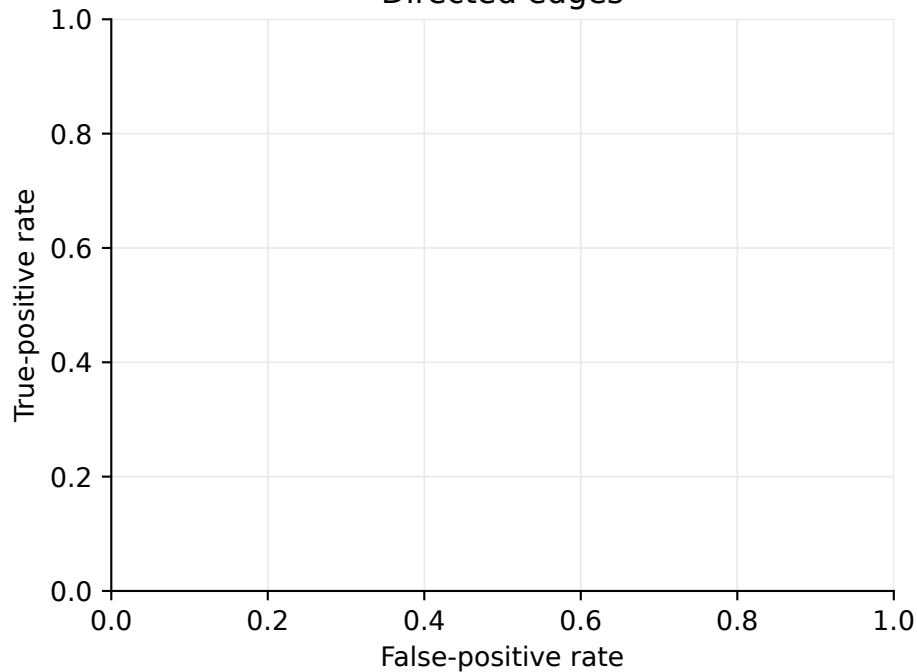
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator collider (scm_5)

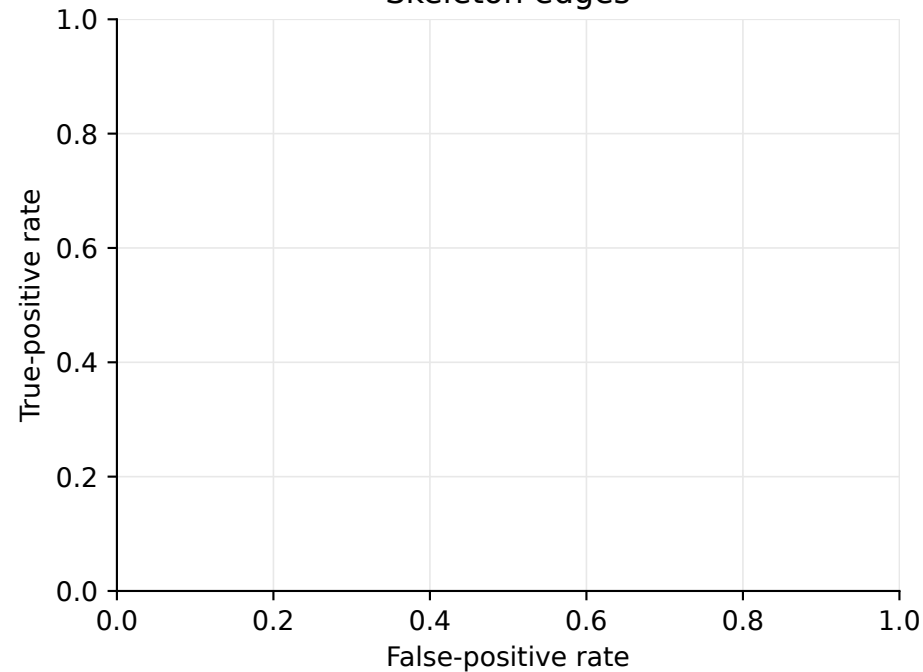
40,000 observations | 10 variables | 23 reference edges | 0/97 successful fits

Full-range normalized paths

Directed edges



Skeleton edges



- PS-MIP: complete targets
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- ◇ PS-MIP: hidden targets (diagnostic)
- △ UT-IGSP: targets supplied
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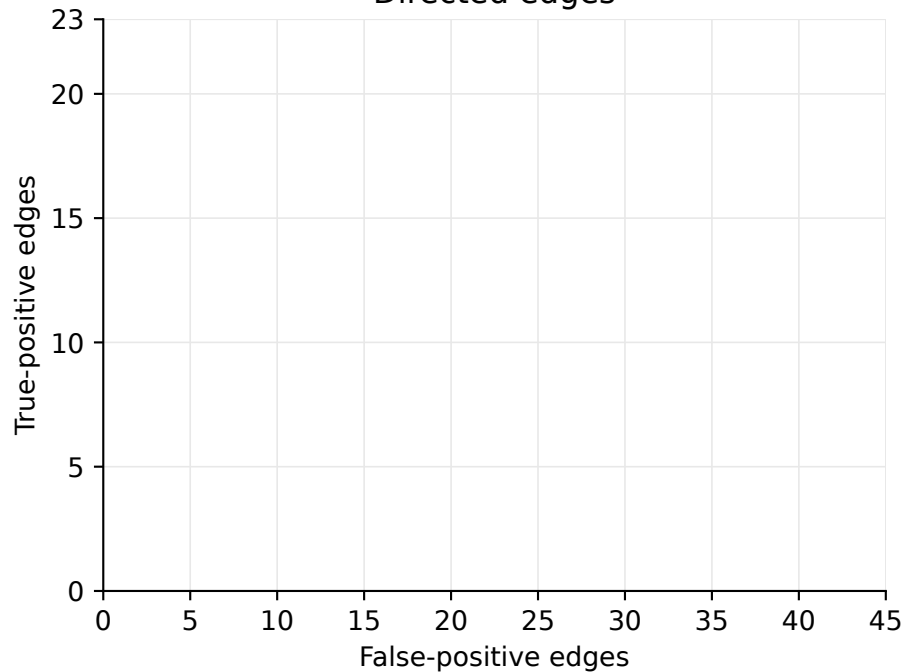
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator collider (scm_5)

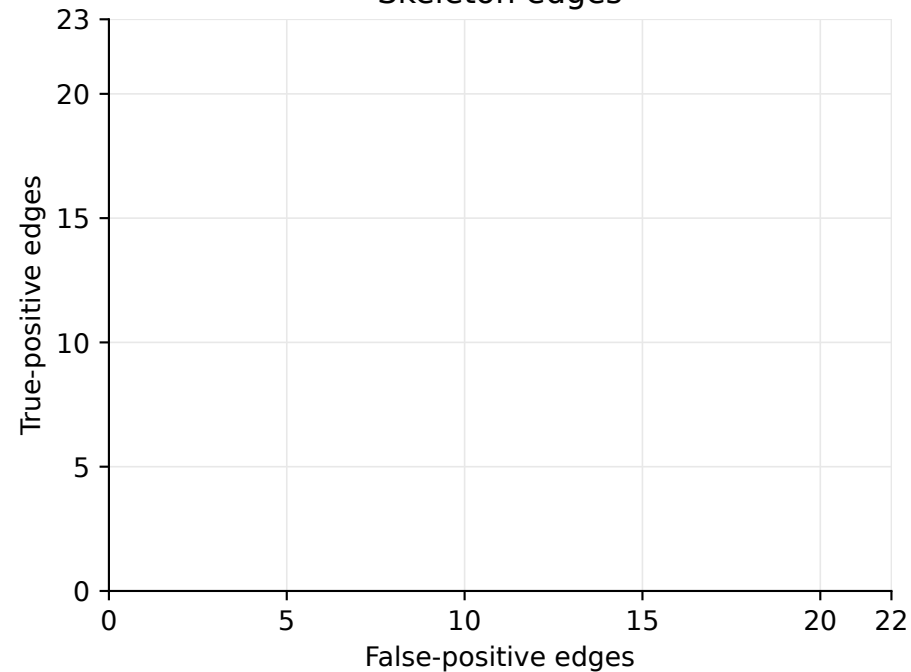
40,000 observations | 10 variables | 23 reference edges | 0/51 successful fits

Full-range paths; every successful operating point retained

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets

- ◇ PS-MIP: hidden targets (diagnostic)

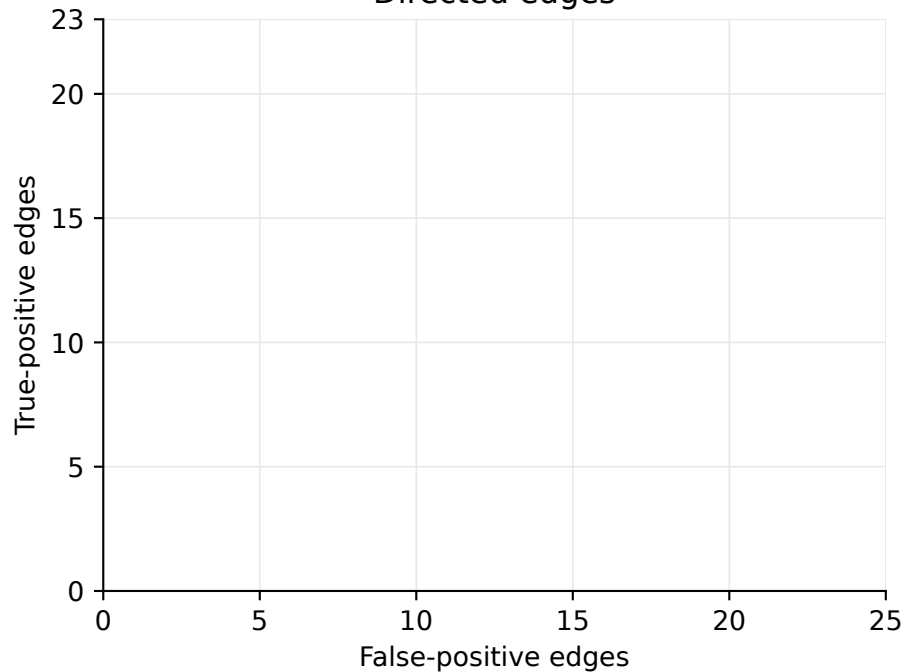
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Causal Chambers | Actuator collider (scm_5)

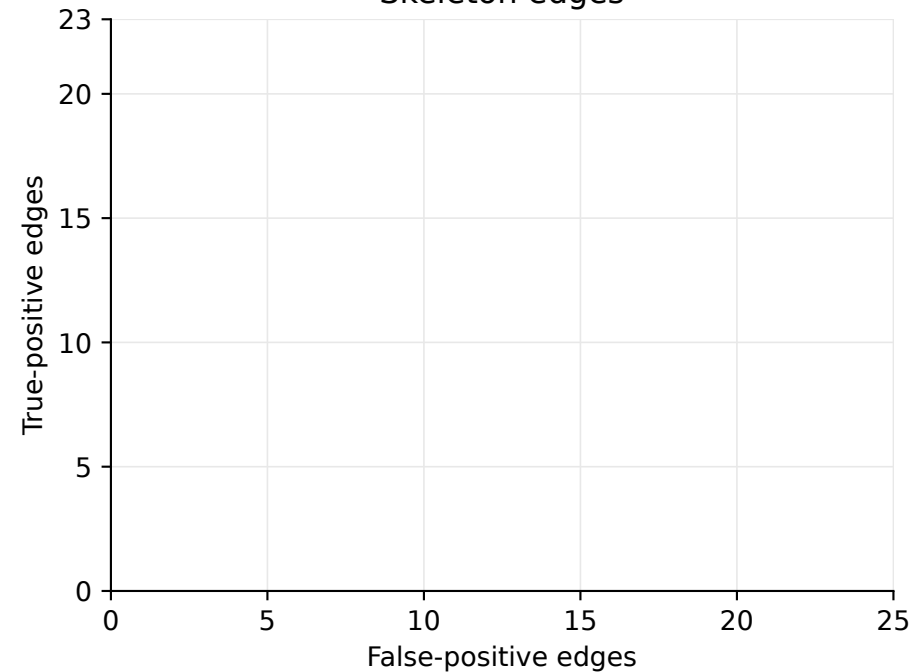
40,000 observations | 10 variables | 23 reference edges | 0/51 successful fits

0-25 FP viewing window; other points remain in full results

Directed edges



Skeleton edges



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- ◇ PS-MIP: hidden targets (diagnostic)

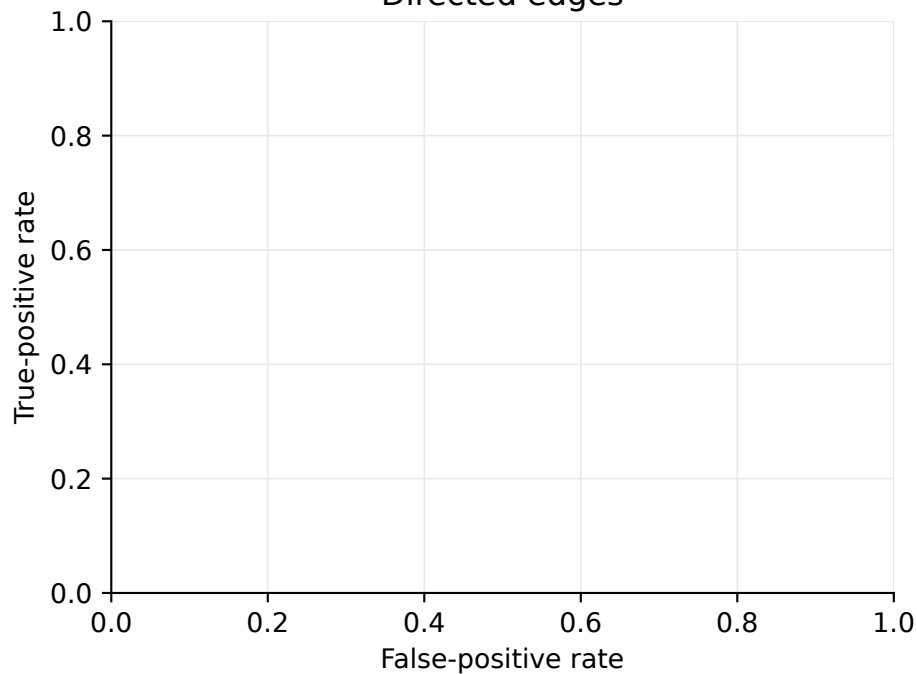
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator collider (scm_5)

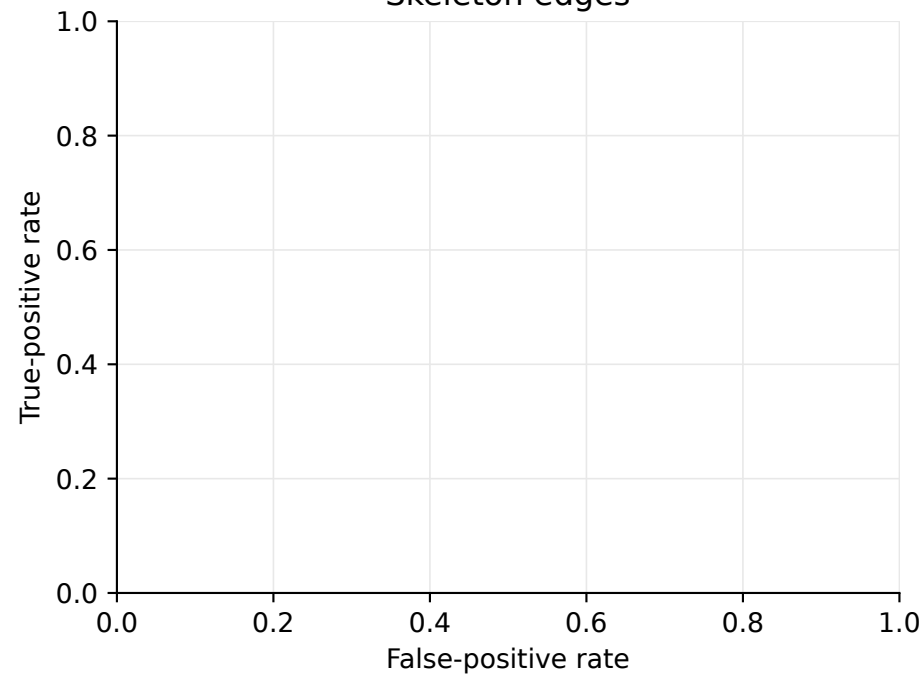
40,000 observations | 10 variables | 23 reference edges | 0/51 successful fits

Full-range normalized paths

Directed edges



Skeleton edges



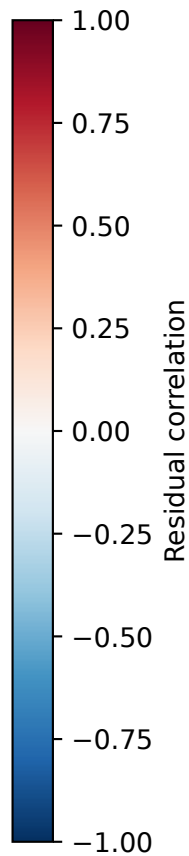
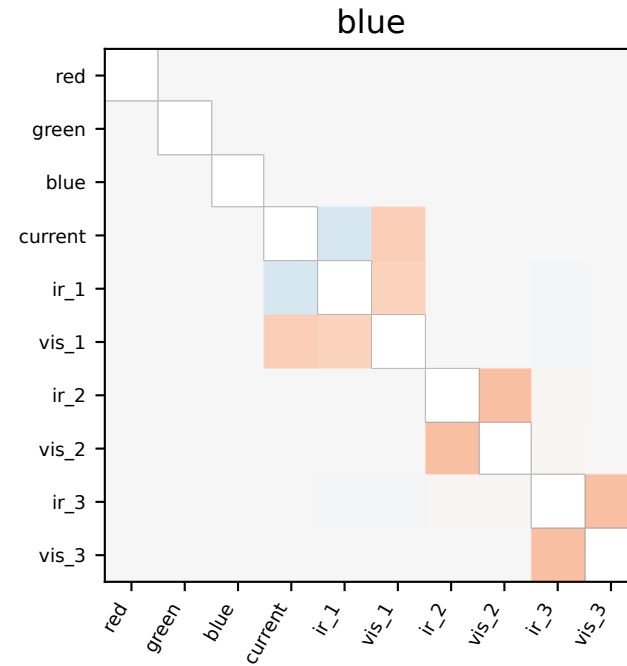
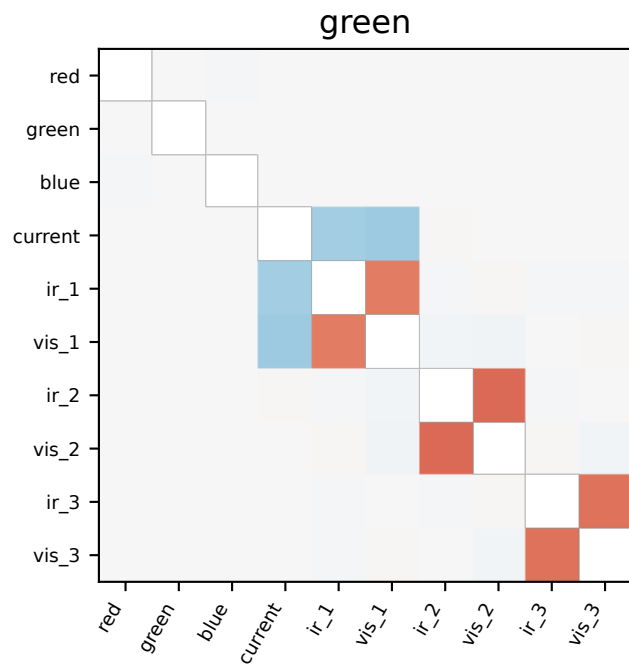
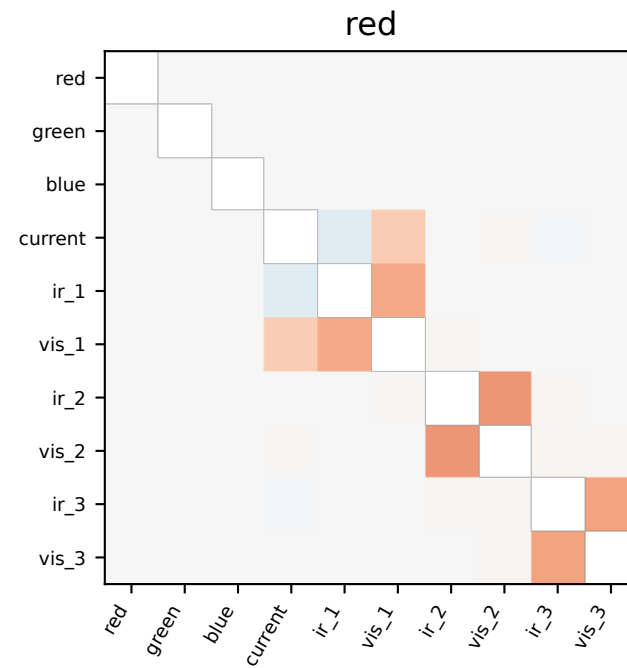
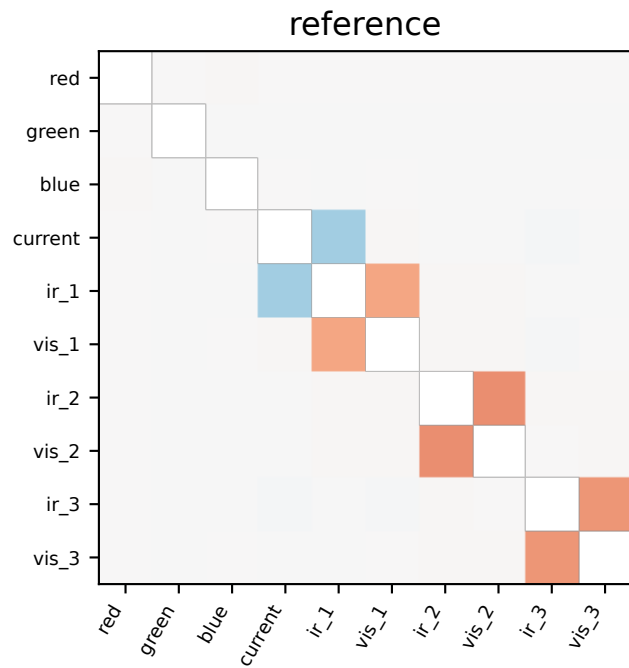
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Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Model check: scm_5

Residuals after regression on documented parents, fitted separately within each environment



Off-diagonal dependence conflicts with independent SEM errors. No rows or variables were removed.
These diagnostics do not identify a unique cause or establish that omitted physical effects are absent.